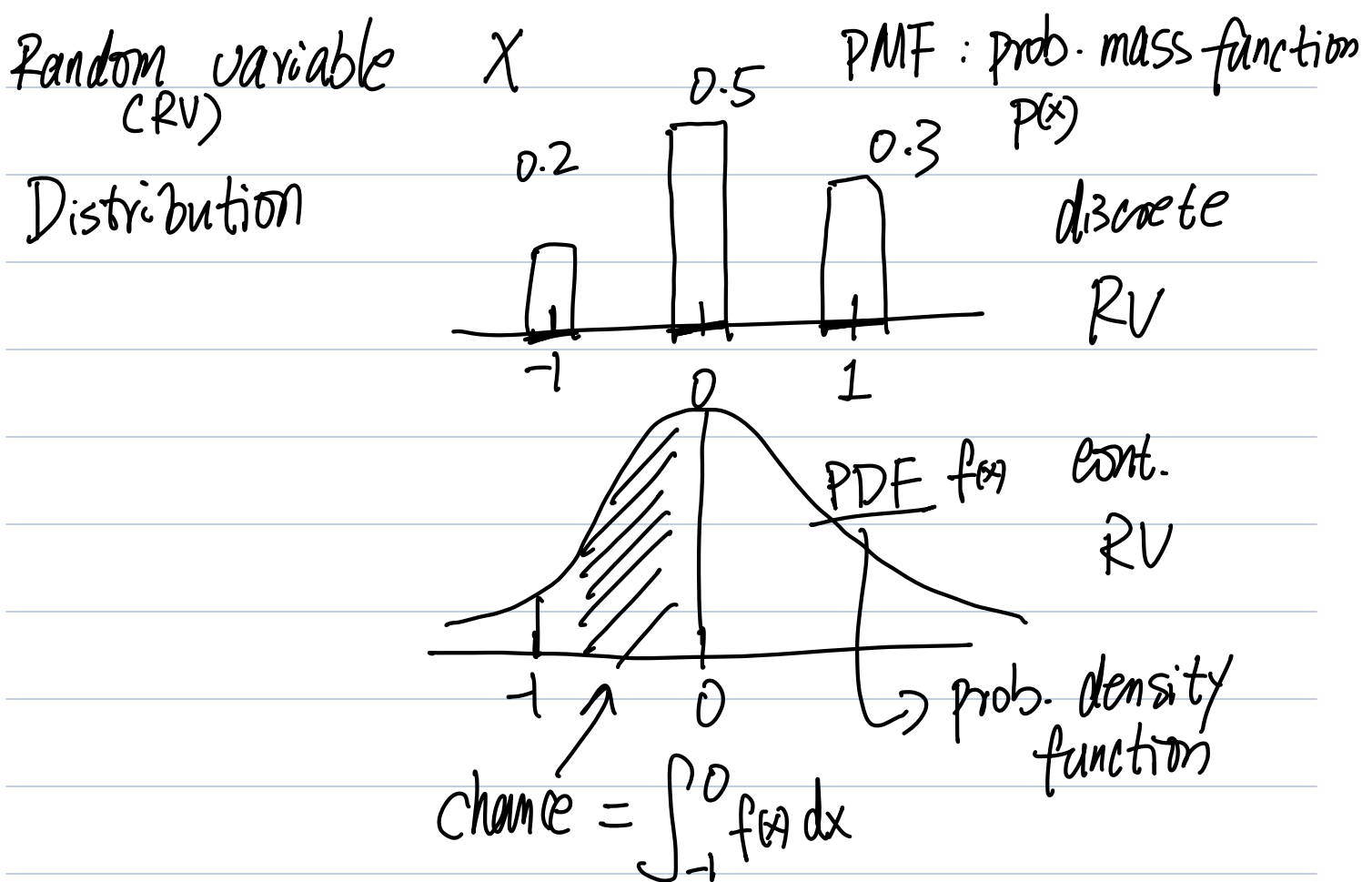




- Expectation (expected value / mean)

$$E[X] = \begin{cases} \sum_x x p(x) & \text{discrete} \\ \int x f(x) dx & \text{continuous} \end{cases}$$

- Variance

$$\text{Var}[X] = E[(X - E[X])^2]$$

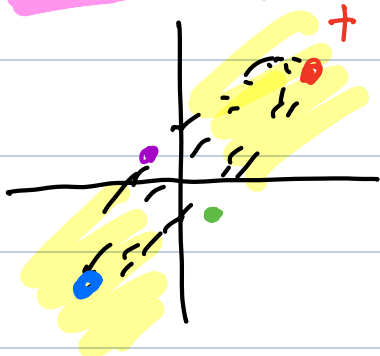
$$= E[X^2] - (E[X])^2$$

☆. Covariance

X, Y : RVs

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Situation 1



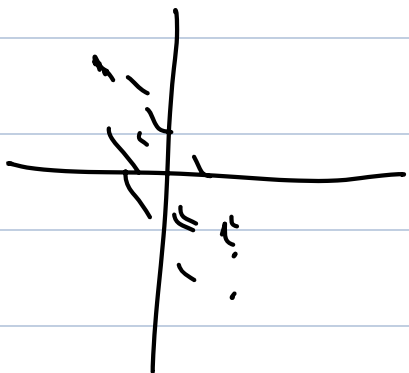
$$+ \times + = +$$

$$- \times - = +$$

$$- \times + = -$$

$$+ \times - = -$$

Situation 2



$$\text{Fact: } \text{Cov}(X, Y) = E[XY] - (E[X] \cdot E[Y])$$

Question: $\text{Cov}(X, X) = ?$

Answer: $\text{Var}(X)$